# ELI — Background Tasks

ELI v2.5.10

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**Updated for v2.4.84.**

## ELI Background Tasks & Unified Code Generation

Two related additions: 1. **All code generation now routes through the verified coding agent** — asking for a script, or self-upgrading, runs the plan → DAG/tree-search → execute → repair → bug-memory pipeline instead of a single LLM shot. 2. **Heavy work runs on background threads** — when a task is estimated heavy (or you ask), ELI starts it on a worker thread, hands back a job id immediately, and you check on it later.

### 1. Unified code generation

| Path | Before | Now |
|---|---|---|
| `CODE_SOLVE` | the coding agent | the coding agent (unchanged) |
| `GENERATE_SCRIPT` | inline single-shot prompt + static checks | **routes through `eli.coding.solve` first** (plan/DAG/tree-search/verify/repair), saving a verified, top-tier script; falls back to the inline generator only if the agent yields nothing or `ELI_GENERATE_SCRIPT_AGENT=0` |
| Self-upgrade (`self_improvement.generate_code_patch`) | LLM patch + import-verify | now also **consults the coding engine's long-term bug memory**: classifies the failure and injects prior fixes for that bug class into the patch prompt |

So "write a script", "generate code", and "improve ELI" all share one engine and one bug-memory — the same consolidation move applied to `recall_memory`. Kill switch: `ELI_GENERATE_SCRIPT_AGENT=0` restores the legacy inline generator.

## 2. Background task manager (`eli/runtime/background_tasks.py`)

In-process, multi-threaded (`ThreadPoolExecutor`). **Distinct** from `eli/planning/jobqueue.py` (which runs durable *external subprocess* jobs) — this runs live in-process Python work (agent runs) for the session.

- `BackgroundTasks.submit(name, fn, *args)` → integer job id; tracks `queued → running → done/failed/cancelled`, result, error, elapsed, note.
- `get(id)`, `list()`, `cancel(id)` (best-effort), `wait(id)`, `stats()`.
- Singleton `get_background_tasks()`; thread-safe.

### When ELI backgrounds a task (`eli/coding/cost.py`)

`should_background(task)` decides: - **Explicit phrasing wins:** "in the background / don't wait / notify me when" → background; "right now / quickly / wait for it" → foreground. - **Otherwise heuristic:** `estimate_cost` scores heavy-compute signals (simulate/monte-carlo/optimise/benchmark/train/dijkstra/parallel…), multi-component structure, length, and planned-step count. `heavy` (≥0.6) ⇒ background.

`_maybe_background_codegen` (executor) applies this at the top of `CODE_SOLVE` and `GENERATE_SCRIPT`: if heavy/explicit, it submits the same action as a background job (re-dispatched with `_no_background` so the worker doesn't re-background) and replies *"started as job #N — say 'check job N'."* Kill switch: `ELI_CODEGEN_BACKGROUND=0`.

### Inspecting jobs

- Actions **`CHECK_JOB`** (`{job_id}` or parsed from "check job 3") and **`BACKGROUND_JOBS`** (list), both in `SUPPORTED_ACTIONS`.
- Router triggers: "check job N" / "job N", "background jobs" / "list jobs".

## Control summary

| Knob | Default | Effect |
|---|---|---|
| `ELI_GENERATE_SCRIPT_AGENT` | on | GENERATE_SCRIPT uses the coding agent (off ⇒ legacy inline) |
| `ELI_CODEGEN_BACKGROUND` | on | auto-background heavy codegen (off ⇒ always foreground) |
| `ELI_AGENT_DAG` / `ELI_CODING_DAG` | on | DAG dispatch / subtask DAG (see `dag.md`) |
| `ELI_CODING_SANDBOX` / `*_RUN_TIMEOUT` / `*_BEAM` / `*_MAX_ITERS` | — | coding-agent execution knobs |

## Scope / honesty

- **Tested deterministically** (`tests/test_background.py`): manager run/fail reporting, cost light-vs-heavy + explicit phrasing, and the CHECK_JOB / BACKGROUND_JOBS executor round-trip. No model needed.
- **Threads, not processes:** background tasks share the process; a running thread can't be force-killed (cancel is best-effort / cooperative). True OS isolation for a job uses the durable subprocess `jobqueue`: run `eli-jobs worker` and queue with `eli-jobs submit --cmd "..."` (jobs survive a restart; one a dead worker left running is marked failed on the next worker start).
- **"LLM deems longer wait":** currently a deterministic estimator (+ explicit phrasing). The planner's decomposition (`plan_steps`) can be fed in for an LLM-informed estimate — hook present, not yet auto-wired.

- **GUI:** the Tasks tab (`eli/gui/tabs/tasks_tab.py`) is the single home for background work: coding jobs and scheduled or overnight tasks (add, edit, cancel). Jobs are also reachable in chat ("check job N", "background jobs") and through the action API.